

KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering
Department of Electrical and Electronic Engineering
Summer – 2024 Mid-term Examination
Program: B.Sc. in EEE (Batch – 17th) 1st Year 2nd Semester
Course Title: Physics - II
Course Code: PHY 0533-1201

PART-A: MCQ

Time: 10 Minutes

Total Marks: 05

Student's ID:

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Date:

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 Invigilator's Signature:

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

CLO1	Define the basic terms related to thermodynamics, modern physics and optics.	4
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Cognitive Remembering

1. How many Isothermal compression steps are present in the Carnot's cycle?
A) 2
B) 3
C) 4
D) 5
E) 6

Cognitive Understanding

2. Most of the natural processes are process.
A) reversible
B) irreversible
C) adiabatic
D) isothermal
E) isobaric

Cognitive Remembering

3. Which one is the correct relation between C_p & C_v ?
A) $C_p > C_v$
B) $C_p = C_v$
C) $C_p < C_v$
D) $C_p + C_v = R$
E) $C_v - C_p = R$

Cognitive Understanding

4. Which of the following is the thermometric properties of thermocouple thermometer?
A) Resistance
B) Volume
C) Electromotive force
D) Brightness
E) Pressure

CLO2	Identify and formulate the various quantities to explain physical and engineering phenomena.	1
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Cognitive Applying

5. The temperature of the heat source of a Carnot's engine is 400K. It takes 840J heat energy from the source at this temperature and rejects heat energy of 630J to the cold sink. What is the temperature of the cold sink?
A) 200K
B) 300K
C) 400K
D) 500K
E) 600K

Signature of Examiner

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PART-B: SAQ

Time: 50 Minutes

Total Marks: 15

[Instruction: Marks for the questions are indicated against each of them]

CLO1	Define the basic terms related to thermodynamics, modern physics and optics.	5
1.	a) State two postulates of the kinetic theory of gas that do not apply to real gases. [2] b) Define: entropy, mechanical equivalent of heat and Internal energy. [3]	
CLO2	Identify and formulate the various quantities to explain physical and engineering phenomena.	5
2.	a) How is energy distributed among the degrees of freedom in a system according to the equipartition of energy? [2] b) Establish a relation between the slopes of adiabatic and isothermal curves. [3]	
CLO3	Apply critical thinking and problem-solving skills to solve a variety of problems concerning thermodynamics, modern physics and optics.	5

[Answer any 01 (one) out of 02 (two) sets of questions]

3. See the stem given below and answer the following questions:

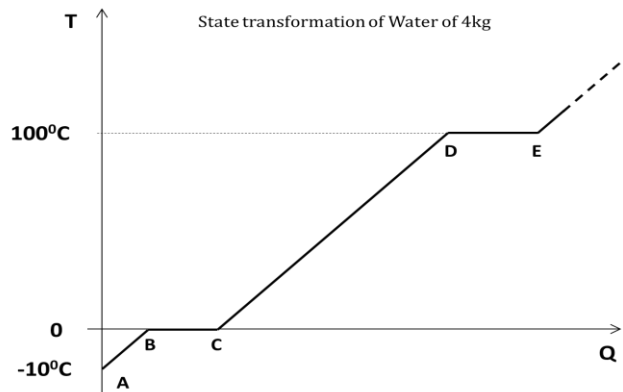


Figure No.-1: State Transformation of 4kg water

- a) Mention the name of the states of A, B, C, D & E. [1]
b) Find the change in entropy for the transformation from A to E. And also compare between BC & DE portion of above figure. [4]
4. a) A platinum resistance thermometer reads 2Ω and 2.73Ω at ice point and steam point respectively. Find the temperature at which it reads 4.83Ω . [1]
b) The mean free path of the molecules of a gas is $6 \times 10^{-8}\text{m}$ and the diameter of a molecule is $2.5 \times 10^{-10}\text{m}$. Find the number of molecules per m^3 . [2]
c) Calculate the values of two molar specific heats for a triatomic ideal gas. [2]